INTERNATIONAL CYBERSECURITY AND DIGITAL FORENSICS ACADEMY

LAB REPORT

PC System Fundamentals, Disk Sectors and Partition Analysis

COURSE	CIP-B101: Basic Computer Skills for Digital Forensics
LAB	Lab 1B: PC System Fundamentals, Disk Sectors and Partition Analysis
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Platform Used	Windows 11 Home (build 26300), HP Laptop 15-fd0xxx, UEFI / GPT
Tools Used	msinfo32, systeminfo, PowerShell (Get-Disk, Get-Partition, Get-Volume), tasklist, Disk Management, DiskPart
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1. Introduction

This lab profiles a Windows 11 laptop the way a junior digital forensic trainee would before an investigation. The system was identified with System Information and systeminfo, the storage layout was read with PowerShell, Disk Management and DiskPart, and running processes were listed with tasklist. Partition sizes were then calculated by hand from sector counts and compared with what the operating system reports. The boot process, MBR/GPT and file systems are explained using evidence taken from the same machine.

The approach is observation only: the aim is to record, calculate and verify without changing anything on the disk.

2. Tools Used and Safety Compliance

Item	Details
Operating system	Windows 11 Home, Version 10.0.26300 (host DESKTOP-0L7FBOA)
Hardware	HP Laptop 15-fd0xxx, Intel Core i5-1334U, 8 GB RAM, 476.94 GB Samsung NVMe SSD
Commands	`msinfo32`, `systeminfo`, `Get-Disk`, `Get-Partition`, `Get-Volume`, `tasklist`, `diskmgmt.msc`, `diskpart` (`list disk`, `select disk 0`, `list partition`)
Calculation	Manual formulas with a calculator, cross-checked with Python 3 (Appendix A)

Part A: System Identification and Evidence Layer Mapping

A1. Basic System Specification

System Information (msinfo32) and systeminfo were captured to record the hardware, firmware and operating system of the machine.

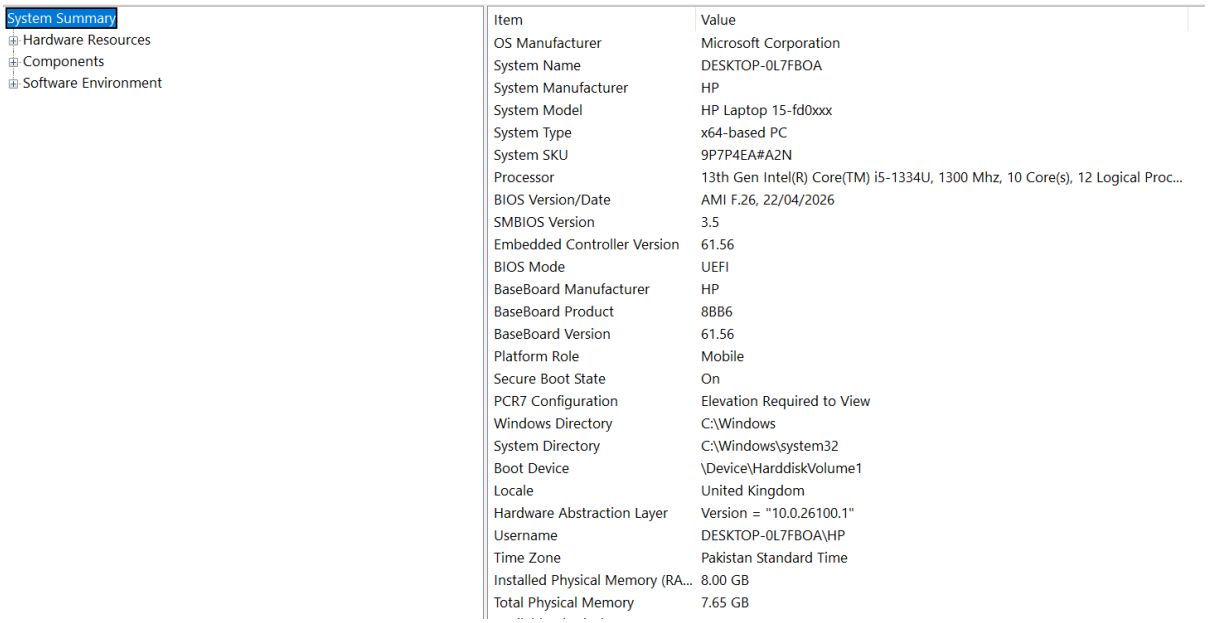


Figure 1: *System Information (msinfo32) summary: model, processor, BIOS mode, Secure Boot state and memory.*



Figure 2: First page of the systeminfo output in Command Prompt.

Item	Value observed	Source
Operating system	Windows 11 Home, Version 10.0.26300, Build 26300	systeminfo
System	HP Laptop 15-fd0xxx, x64-based PC, platform role Mobile	msinfo32
Processor	13th Gen Intel Core i5-1334U, 1300 MHz, 10 cores, 12 logical processors	msinfo32
Memory	8.00 GB installed, 7.65 GB usable (7,836 MB total, 814 MB free at capture)	msinfo32, systeminfo
BIOS	AMI F.26, dated 22/04/2026	msinfo32, systeminfo
BIOS mode / Secure Boot	UEFI / On	msinfo32
Install date / time zone	02/02/2025, 6:48:16 pm / (UTC+05:00) Islamabad, Karachi	systeminfo

The OS Name and Version rows are outside the msinfo32 view, so they are evidenced by systeminfo. The install date, boot time and time zone anchor a timeline, and UEFI with Secure Boot already points to a GPT disk and EFI boot, which Parts D and E confirm.

A2. Storage Information

wmic was not used; the PowerShell alternative allowed by the lab was run instead.



Figure 3: Get-Disk: one disk, Healthy, Online, 476.94 GB, GPT partition style.

```
PS C:\Users\Computer Arena> Get-Partition


   DiskPath: \\?\scsi#disk&ven_nvme&prod_samsung_mzvl4512#5&d12030e&0&000000#{53f56307-b6bf-11d0-94f2-00a0c91efb8b}

PartitionNumber  DriveLetter Offset                                        Size Type
---------------  ----------- ------                                        ---- ----
1                            1048576                                    100 MB System
2                            105906176                                   16 MB Reserved
3                C           122683392                                 116.35 GB Basic
4                            125055270912                                738 MB Recovery
5                D           125830168576                              359.75 GB Basic


PS C:\Users\Computer Arena>
```

Figure 4: Get-Partition: five partitions on Disk 0 with drive letters, offsets, sizes and types.

```
PS C:\Users\Computer Arena> Get-Volume

DriveLetter FriendlyName FileSystemType DriveType HealthStatus OperationalStatus SizeRemaining      Size
----------- ------------ -------------- --------- ------------ ----------------- -------------      ----
D                        NTFS           Fixed     Healthy      OK                    166.74 GB 359.75 GB
C                        NTFS           Fixed     Healthy      OK                     12.09 GB 116.35 GB
                         NTFS           Fixed     Healthy      OK                     53.38 MB    738 MB


PS C:\Users\Computer Arena>
```

Figure 5: Get-Volume: volumes C: and D: and an unlettered recovery volume, all NTFS and Healthy.

Disk 0 is a Samsung NVMe SSD (model string MZVL4512) of 476.94 GB that uses GPT. It holds five partitions: System (100 MB), Reserved (16 MB), C: (116.35 GB), Recovery (738 MB) and D: (359.75 GB). Get-Volume reports all three data volumes as NTFS, with C: at 12.09 GB free of 116.35 GB and D: at 166.74 GB free of 359.75 GB.

A3. Evidence Layer Mapping

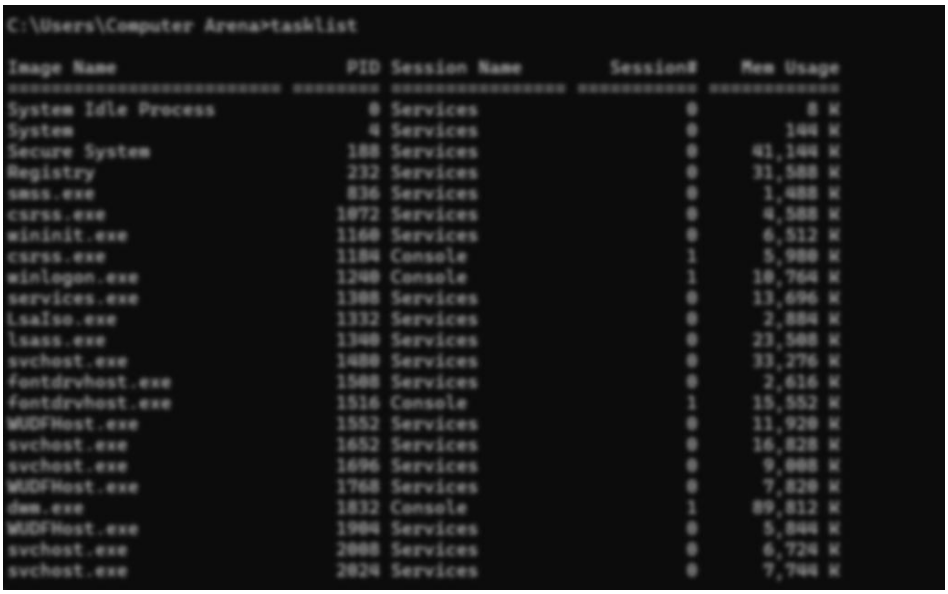
User activity is produced by applications, the operating system and the disk or memory underneath them. Each layer keeps its own traces.

Layer / Source	Example evidence	Where it may be found	Why it matters
User layer	Opened files, searched terms, typed usernames	Recent files, browser history, shell history	Links a user action to system activity
Application layer	Office document metadata, browser cache, chat logs	Application folders and databases	Shows what software created or used the evidence
OS layer	Event logs, installed programs, user profiles	Windows logs, registry, Linux `/var/log`	Shows system events and account activity
Disk layer	Deleted files, file fragments, partitions	File system and unallocated space	Supports recovery and timeline analysis
Memory / process layer	Running malware, PIDs, network connections	RAM capture, process list	Shows live system behaviour
Firmware / hardware layer	BIOS/UEFI version, boot mode, Secure Boot state	msinfo32, systeminfo, firmware settings	Explains boot behaviour and hardware identity

Why evidence exists in more than one layer: one user action leaves traces in several layers at once. Opening a downloaded document, for example, appears in the application (recent-file list, browser history), the operating system (event logs, registry), on disk (a file-system entry with timestamps) and in memory (a running process). If one trace is deleted or altered, the others can still confirm what happened, and agreement between layers makes a finding more reliable.

Part B: Program, Process and Thread Investigation

B1. Running Processes



```
C:\Users\Computer Arena>tasklist

Image Name                     PID Session Name        Session#    Mem Usage
========================= ======== ================ =========== ============
System Idle Process              0 Services                   0          8 K
System                           4 Services                   0        144 K
Secure System                  188 Services                   0     41,144 K
Registry                       232 Services                   0     31,588 K
smss.exe                       836 Services                   0      1,488 K
csrss.exe                     1072 Services                   0      4,588 K
wininit.exe                   1160 Services                   0      6,512 K
csrss.exe                     1184 Console                    1      5,988 K
winlogon.exe                  1240 Console                    1     10,764 K
services.exe                  1308 Services                   0     13,696 K
LsaIso.exe                    1332 Services                   0      2,884 K
lsass.exe                     1340 Services                   0     23,508 K
svchost.exe                   1480 Services                   0     33,276 K
fontdrvhost.exe               1508 Services                   0      2,616 K
fontdrvhost.exe               1516 Console                    1     15,552 K
WUDFHost.exe                  1552 Services                   0     11,920 K
svchost.exe                   1652 Services                   0     16,828 K
svchost.exe                   1696 Services                   0      9,008 K
WUDFHost.exe                  1768 Services                   0      7,820 K
dwm.exe                       1832 Console                    1     89,812 K
WUDFHost.exe                  1904 Services                   0      5,844 K
svchost.exe                   2008 Services                   0      6,724 K
svchost.exe                   2024 Services                   0      7,744 K
```

Figure 6: *tasklist output (first page) showing process names, PIDs, sessions and memory usage.*

Process	PID	Session	Mem usage	Likely purpose
winlogon.exe	1240	Console (1)	10,764 K	Manages user logon and logoff and the secure sign-in screen.
lsass.exe	1340	Services (0)	23,508 K	Local Security Authority: enforces security policy and handles authentication. Often targeted for credential theft, so it is checked in malware triage.
dwm.exe	1832	Console (1)	89,812 K	Desktop Window Manager: draws and composites the windows on screen. It has the highest memory use in the visible list.

Background services run in session 0, while interactive processes such as winlogon.exe and dwm.exe run in session 1 (Console).

B2. Program, Process and Thread

- **Program:** a passive file on disk, such as an `.exe`, that holds instructions but is not running.
- **Process:** a running instance of a program with its own PID, memory space and handles, for example dwm.exe with PID 1832.
- **Thread:** the smallest unit of execution inside a process. Threads of one process share its memory, but each has its own stack and registers.

Forensic use: a process list shows what was running at that moment. Unexpected names, a wrong session or parent, look-alike names such as `svch0st.exe` or unusual memory use can point to malware. PIDs link a process to its network connections, open files and memory. The list is lost at shutdown, so it has to be captured early in live triage.

Part C: Disk Sector and Disk Size Calculation

> **Key formula**
> Disk or partition size = number of sectors × bytes per sector. To convert bytes to GB, divide by 1024 × 1024 × 1024. In this report GB follows the lab convention of 1024^3 bytes.

C1. Slide Calculation

Given: sectors per track = 400, heads = 12, cylinders = 17,000, bytes per sector = 512.

Item	Formula	Answer
Total sectors	400 × 12 × 17,000	81,600,000 sectors
Total bytes	81,600,000 × 512	41,779,200,000 bytes
Kilobytes	41,779,200,000 ÷ 1024	40,800,000 KB
Megabytes	40,800,000 ÷ 1024	39,843.75 MB
Gigabytes	39,843.75 ÷ 1024	38.91 GB

C2. Partition Size Calculation on My Disk

Figure 7 gives the exact byte values from the partition table. A sector size of 512 bytes was used, as in the lab formula; every offset is a multiple of 512, and the 1 MB (1,048,576 byte) offset of partition 1 equals 2,048 sectors. Start sector = offset ÷ 512 and sector count = size ÷ 512.

```
PS C:\Users\Computer Arena> Get-Partition | Format-Table DiskNumber,PartitionNumber,DriveLetter,Offset,Size,Type

DiskNumber PartitionNumber DriveLetter       Offset         Size Type
---------- --------------- -----------       ------         ---- ----
         0               1                  1048576    104857600 System
         0               2                105906176     16777216 Reserved
         0               3           C    122683392 124932587520 Basic
         0               4           125055270912    773849088 Recovery
         0               5           D 125830168576 386279669760 Basic


PS C:\Users\Computer Arena>
```

Figure 7: Get-Partition | Format-Table: exact offsets and sizes in bytes for all five partitions.

Partition	Type	Offset (bytes)	Start sector	Sector count	End sector
1	System	1,048,576	2,048	204,800	206,847
2	Reserved	105,906,176	206,848	32,768	239,615
3 (C:)	Basic	122,683,392	239,616	244,008,960	244,248,575
4	Recovery	125,055,270,912	244,248,576	1,511,424	245,759,999
5 (D:)	Basic	125,830,168,576	245,762,048	754,452,480	1,000,214,527

Size = sector count × 512 ÷ 1024 ÷ 1024 ÷ 1024:

Partition	Working	Size (GB)
1	204,800 × 512 = 104,857,600 bytes ÷ 1,073,741,824	0.0977 (100 MB)
2	32,768 × 512 = 16,777,216 bytes ÷ 1,073,741,824	0.0156 (16 MB)
3 (C:)	244,008,960 × 512 = 124,932,587,520 bytes ÷ 1,073,741,824	116.35
4	1,511,424 × 512 = 773,849,088 bytes ÷ 1,073,741,824	0.7207 (738 MB)
5 (D:)	754,452,480 × 512 = 386,279,669,760 bytes ÷ 1,073,741,824	359.75

Sum check: 0.0977 + 0.0156 + 116.3525 + 0.7207 + 359.7510 = 476.9375 GB, which rounds to the 476.94 GB reported by Get-Disk. **Continuity check:** each partition starts where the previous one ends (2,048 + 204,800 = 206,848 and 206,848 + 32,768 = 239,616). A 2,048-sector (1 MB) alignment gap separates partition 4 from partition 5.

Part D: Safe Partition Table Inspection

D1. Disk Management

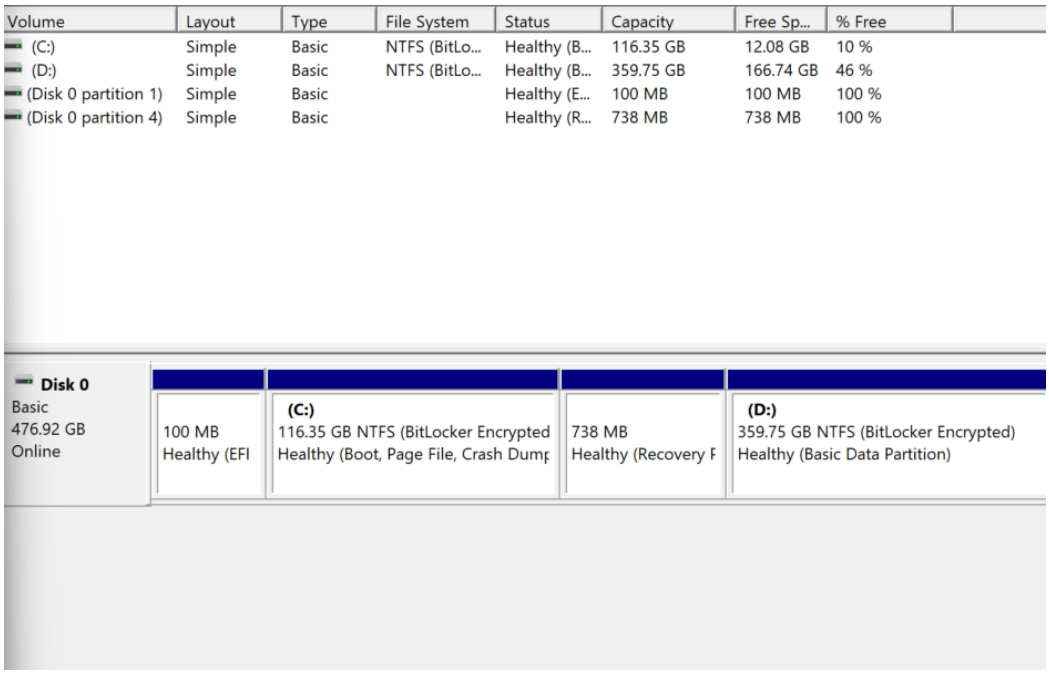


Figure 8: *Disk Management: Disk 0 layout with EFI, C:, Recovery and D: partitions.*

Disk Management shows Disk 0 as Basic, 476.92 GB and Online. The layout is a 100 MB EFI partition, C: (116.35 GB, NTFS, BitLocker Encrypted, Boot, Page File, Crash Dump), a 738 MB Recovery partition and D: (359.75 GB, NTFS, BitLocker Encrypted). The 16 MB Reserved partition is not displayed, because Disk Management hides it.

D2. PowerShell

Get-Disk, Get-Partition and Get-Volume were run as a normal user (Figures 3 to 5), plus a Format-Table view for exact byte values (Figure 7). They confirm the GPT disk and the five partitions listed in Part C.

D3. DiskPart (listing only)

```
icrosoft DiskPart version 10.0.26100.1150

opyright (C) Microsoft Corporation.
n computer: DESKTOP-0L7FBOA

ISKPART> list disk

 Disk ###  Status         Size     Free     Dyn  Gpt
 --------  -------------  -------  -------  ---  ---
 Disk 0    Online          476 GB  1024 KB        *

ISKPART>
```

Figure 9: *DiskPart: list disk shows Disk 0 Online, 476 GB, with the GPT column marked (*).*

```
DISKPART> select disk 0

Disk 0 is now the selected disk.

DISKPART> list partition

  Partition ###  Type              Size     Offset
  -------------  ----------------  -------  -------
  Partition 1    System             100 MB  1024 KB
  Partition 2    Reserved            16 MB   101 MB
  Partition 3    Primary            116 GB   117 MB
  Partition 4    Recovery           738 MB   116 GB
  Partition 5    Primary            359 GB   117 GB

DISKPART>
```

Figure 10: *DiskPart: select disk 0 and list partition for the five partitions.*

DiskPart agrees with the other tools: the GPT column is marked and the partition types, sizes and offsets match Figure 7 (for example partition 1 at offset 1024 KB = 1,048,576 bytes). DiskPart truncates sizes instead of rounding (476 GB, 116 GB, 359 GB), and its 1024 KB of free space matches the 1 MB gap found in Part C.

The optional hex-viewer step (WinHex, sector 0) was not performed; the partition style was confirmed with Get-Disk and DiskPart instead.

Part E: Boot Process, MBR and GPT

E1. PC Boot Process

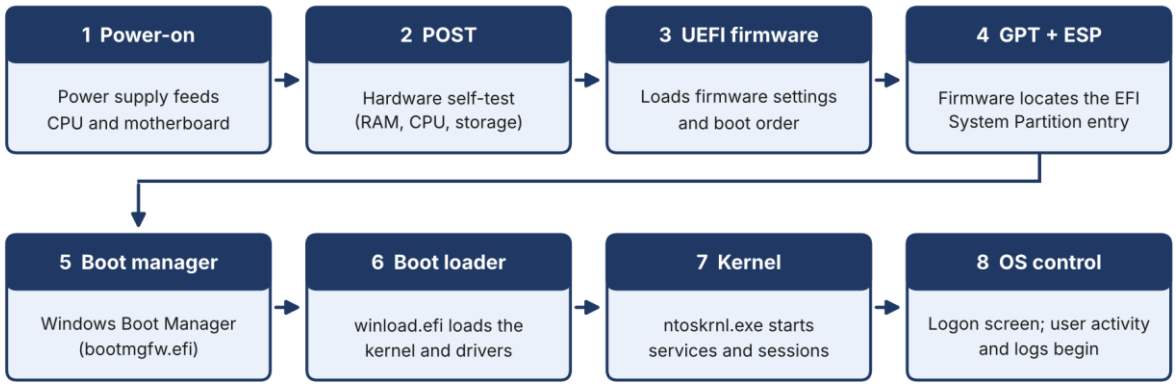


Figure 11: Boot sequence of this UEFI/GPT system, from power-on to operating system control.

Stage	What happens	Forensic relevance
Power-on and POST	Hardware is powered and self-tested.	Hardware failure or tampering may affect evidence access.
UEFI firmware (CMOS/NVRAM settings)	Firmware settings and boot device order are applied.	Boot mode explains the partition style and boot artefacts.
GPT / EFI boot record	Firmware finds the EFI System Partition and its boot entry.	Boot records can show the boot configuration.
Boot manager and loader	Windows Boot Manager runs, then winload.efi loads the kernel and drivers.	May show multi-boot or hidden OS behaviour.
Kernel and OS control	ntoskrnl.exe starts services and the logon screen appears.	Logs and user activity begin to accumulate.

Boot mode of this system: UEFI

msinfo32 reports BIOS Mode = UEFI with Secure Boot On (Figure 1), and Get-Disk and DiskPart show a GPT disk (Figures 3 and 9). The machine therefore boots through the EFI System Partition (partition 1, 100 MB) and not through an MBR boot sector.

E2. MBR versus GPT

Feature	MBR	GPT
Age / design	Older partitioning system from the 1980s, used with legacy BIOS.	Modern system used with UEFI.
Size limit	Commonly about 2 TB (32-bit sector addresses).	Very large disks; the slide notes up to about 9.4 ZB.
Partition information	Partition table and boot code sit in the first sector (sector 0).	Protective MBR in sector 0, GPT header in sector 1, partition entries after it, and a backup copy at the end of the disk.
Forensic issue	Damage to sector 0 can break partition interpretation.	Backup structures help recovery and validation; the protective MBR can mislead tools that only read sector 0.
On this system	Not used.	Used: Disk 0 is GPT (Figures 3 and 9).

Part F: File System Identification

Volume	Partition	File system	Evidence
C:	3 (116.35 GB)	NTFS	Get-Volume (Figure 5); Disk Management: NTFS, BitLocker Encrypted (Figure 8)
D:	5 (359.75 GB)	NTFS	Get-Volume (Figure 5); Disk Management: NTFS, BitLocker Encrypted (Figure 8)
No letter	4 (738 MB)	NTFS	Get-Volume lists an unlettered 738 MB NTFS volume (Figure 5); Recovery partition in Figure 8
No letter	1 (100 MB)	Not displayed	EFI System Partition; the tools show no file system. By UEFI design it is FAT-based, which is expected but was not observed.

File system	Common OS / usage	Forensic importance	Found here?
FAT / FAT32	Removable media, older systems	Simple structure; often used on USB drives.	Expected on EFI partition
NTFS	Modern Windows	MFT entries, timestamps, permissions and journals.	Yes: C:, D:, Recovery
ext family	Linux	Inode-based metadata; supports timeline analysis.	No
XFS / JFS / btrfs	Linux / Unix-like	Advanced features can affect recovery and metadata interpretation.	No
HFS+	Older macOS	Important in Mac investigations; limited date support.	No

Why the file system matters: it defines how files, names, timestamps and permissions are stored on a partition. It lets an examiner locate files, recover deleted entries, read timestamps and build timelines; NTFS keeps this in the MFT and its journals. Identifying it first also decides which tools and recovery methods apply.

> **Observation: BitLocker**
> Disk Management marks both C: and D: as BitLocker Encrypted. In a real case these volumes could not be read from a powered-off disk image without the recovery key, so live acquisition or key recovery would have to be planned first.

Part G: Final Practical Assignment (Partitions 1 and 2)

G1. Verification Steps

1. Ran `Get-Partition | Format-Table DiskNumber,PartitionNumber,DriveLetter,Offset,Size,Type` to get the exact offset and size in bytes (Figure 7).
2. Converted each offset to a start sector: ÷ 512 (partition 1: 1,048,576 ÷ 512 = 2,048; partition 2: 105,906,176 ÷ 512 = 206,848).
3. Converted each size to a sector count: ÷ 512 (partition 1: 104,857,600 ÷ 512 = 204,800; partition 2: 16,777,216 ÷ 512 = 32,768).
4. Calculated each size in GB: sector count × 512 ÷ 1024^3.
5. Checked that partition 1 ends at sector 206,847, so partition 2 starts at 206,848 with no gap.
6. Compared the results with Get-Partition, DiskPart and Disk Management (Figures 4, 8, 9, 10).

G2. Results

Task	Evidence	Answer
Identify partition 1	Figures 4, 7, 10	Type System (EFI System Partition), offset 1,048,576 bytes, 100 MB
Partition 1 sectors	Figure 7, Part C2	Start 2,048; count 204,800; end 206,847
Partition 1 size	204,800 × 512 ÷ 1024^3	0.0977 GB (100 MB)
Identify partition 2	Figures 4, 7, 10	Type Reserved (Microsoft Reserved), offset 105,906,176 bytes, 16 MB
Partition 2 sectors	Figure 7, Part C2	Start 206,848; count 32,768; end 239,615
Partition 2 size	32,768 × 512 ÷ 1024^3	0.0156 GB (16 MB)

Partition	Calculated	Get-Partition	DiskPart	Disk Mgmt	Match
1 (System)	0.0977 GB	100 MB	100 MB	100 MB	Yes
2 (Reserved)	0.0156 GB	16 MB	16 MB	Not shown	Yes

Difference explained: the calculated and reported values are the same size in different units. 0.0977 GB is 100 MB (100 MiB) because the tools use 1024-based units but label them MB and GB. DiskPart truncates and the other tools round. The same GB/GiB effect explains why the drive is sold as 512 GB (decimal) but shows as 476.94 GB. Partitions 1 and 2 have no drive letter or volume, so there is no file-system overhead to account for. Disk Management does not display partition 2 because it hides the Microsoft Reserved partition.

Screenshot Checklist

No.	Screenshot evidence	Included?	Figure
1	System Information or systeminfo output	Yes	1, 2
2	Disk drive model and size output	Yes	3, 4
3	Logical volume / file system output	Yes	5
4	Running process / PID output	Yes	6
5	Disk Management, Get-Partition, DiskPart output	Yes	4, 7, 8, 9, 10
6	Partition 1 calculation evidence	Yes	7, Part C2, G
7	Partition 2 calculation evidence	Yes	7, Part C2, G
8	Boot mode evidence: BIOS/UEFI	Yes	1, 9
9	File system identification evidence	Yes	5, 8
10	Final report summary / conclusion page	Yes	Conclusion

Conclusion

This lab profiled a Windows 11 laptop without changing it. The system runs in UEFI mode with Secure Boot on and boots from a GPT disk holding five partitions. By converting offsets and sizes from the partition table into sectors, I calculated partition 1 as 0.0977 GB (204,800 sectors) and partition 2 as 0.0156 GB (32,768 sectors); the five partitions add up to 476.94 GB, matching Get-Disk, and the figures agree with DiskPart and Disk Management. C:, D: and the recovery volume use NTFS, and C: and D: are BitLocker encrypted.

These skills matter in digital forensics because evidence is only trustworthy when it can be located, measured and verified with read-only methods. Knowing the layers, processes, partition layout, boot process and file system tells an examiner where to look, which tools to use and what could be missed, such as hidden partitions or encrypted volumes.

Appendix A: Python Cross-check of Calculations

The manual calculations were re-computed with a short Python script (read-only arithmetic on the values from Figure 7). Output columns: partition, start sector, sector count, end sector, size. The results match Part C.

```
G = 1024**3
parts = [(1,1048576,104857600), (2,105906176,16777216),
         (3,122683392,124932587520), (4,125055270912,773849088),
         (5,125830168576,386279669760)]
total = 0
for n, offset, size in parts:
    start, count = offset // 512, size // 512
    total += size
    print(n, f"{start:,}", f"{count:,}", f"{start+count-1:,}", f"{count*512/G:.4f} GB")
print("Total", f"{total/G:.4f} GB")

1 2,048 204,800 206,847 0.0977 GB
2 206,848 32,768 239,615 0.0156 GB
3 239,616 244,008,960 244,248,575 116.3525 GB
4 244,248,576 1,511,424 245,759,999 0.7207 GB
5 245,762,048 754,452,480 1,000,214,527 359.7510 GB
Total 476.9375 GB
```